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Impact of environmental pesticide exposure on hematological and biochemical health parameters in free-grazing cattle of Saharsa District, Bihar

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Abstract- The indiscriminate use of synthetic chemical pesticides-particularly organophosphates, carbamates, and pyrethroids-has become a hallmark of modern Indian agriculture. While effective in crop protection, these chemicals pose significant risks to non-target organisms, including livestock. This study was conducted in six pesticide-intensive villages of Saharsa district, Bihar, to evaluate the physiological and biochemical effects of environmental pesticide exposure in cattle (*Bos indicus*). Sixty adult cattle were divided into two groups: an exposed group (n=30) that grazed near pesticide-treated fields, and a control group (n=30) maintained on stall-fed diets with no field exposure. Clinical examinations, hematological profiling (Hb, RBC, WBC, PCV), and biochemical assays (ALT, AST, creatinine, urea, total protein) were conducted to assess health impacts. Cattle in the exposed group exhibited clear signs of subclinical toxicity, including hypersalivation, diarrhea, dull hair coat, reduced milk yield, and lethargy. Hematological analysis revealed significant reductions in hemoglobin, RBC count, and PCV, indicating anemia, while WBC counts were elevated, suggesting systemic inflammation. Biochemical results showed markedly increased levels of ALT and AST, indicating hepatic stress, alongside elevated creatinine and urea levels, reflecting early renal dysfunction. A modest decline in total protein levels suggested metabolic or nutritional compromise. These findings demonstrate that chronic exposure to agricultural pesticides can significantly impair cattle health, particularly in open-grazing systems. The study underscores the need for safer pesticide practices, controlled grazing, and farmer education to mitigate the unintended consequences of chemical agriculture on livestock. Further research is rezcommended to explore long-term effects and potential mitigation strategies.

Key words: Cattle health, pesticide toxicity, hematology, non-target effects, organophosphates, agriculture

INTRODUCTION

Widespread use of synthetic chemical pesticides namely organophosphates, carbamates, and pyrethroids is a common practice in Indian agriculture to manage pest infestations and enhance crop yield. While these agrochemicals are effective in protecting crops, their broader ecological and health-related consequences often go unaddressed. In rural regions like Saharsa, Bihar, it is a customary practice to allow cattle to graze freely on or near agricultural lands.[1] This proximity to cultivated fields exposes the animals to residual pesticides present in fodder, water sources, and soil. Exposure can occur through ingestion of contaminated plant material, consumption of polluted water, or even direct skin contact. Prolonged or even short-term exposure to such chemicals can have deleterious effects on the animals' physiology- particularly impairing liver function, disrupting nervous system activity, and affecting reproductive health.[2-4] Given the increasing dependency on chemical pesticides and the potential risks to livestock health, the present study is designed to

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investigate and evaluate the physiological and biochemical alterations in cattle resulting from pesticide exposure. Specifically, the research focuses on cattle grazing in intensively farmed areas of the Saharsa district, where pesticide use is prevalent. Through this study, we aim to provide insights that could inform safer agricultural practices and promote livestock health management in pesticide-exposed environments.

MATERIALS & METHODS

Study Area and Animal Selection

The study was carried out in six representative villages of Saharsa district, Bihar, India (Latitude: 25°52'N, Longitude: 86°36'E), a region characterized by intensive agricultural activity, particularly paddy and vegetable cultivation. These crops are commonly treated with a wide range of chemical pesticides including organophosphates, carbamates, and pyrethroids, which have the potential to contaminate surrounding ecosystems.

To evaluate the impact of environmental pesticide exposure on bovine health, a total of 60 adult cattle (*Bos indicus*) of similar age and physiological condition were selected and divided into two experimental groups based on their management and grazing patterns:

Exposed Group (n = 30): These were free-grazing cattle that routinely accessed agricultural fields known to be treated with chemical pesticides. Animals in this group were selected from households that allowed open grazing adjacent to or within farmlands during and after pesticide application periods.

Control Group (n = 30): This group consisted of stall-fed cattle reared under semi-intensive systems, without access to agricultural fields. These animals were fed with stored fodder, minimizing any potential exposure to pesticides.

A preliminary health assessment was conducted through visual and physical examination of all selected cattle. Clinical signs indicative of possible toxic exposure were recorded, including:

- Excessive salivation
- Watery diarrhea
- Rough or dull hair coat
- Changes in appetite and behavior
- Variations in milk production (in lactating cows)

Each symptom was noted to establish a baseline for comparison between the exposed and control groups.

Blood Sample Collection

From each animal, approximately 5–7 ml of blood was collected via jugular venipuncture using sterile disposable syringes and vacutainers. The blood samples were divided into:

- EDTA-containing tubes for hematological analysis
- Plain tubes for serum separation and biochemical analysis

Samples were immediately transported in ice-cooled containers to the laboratory for analysis.

Hematological Analysis

Standard veterinary diagnostic protocols were followed to analyze the following hematological parameters:

- Hemoglobin (Hb) content (g/dL)
- Red Blood Cell (RBC) count (×10v /μ L)
- White Blood Cell (WBC) count (×10^3/μ L)
- Packed Cell Volume (PCV) (%)

These parameters were measured using automated hematology analyzers and manual techniques (where required), adhering to the guidelines provided by the Clinical and Laboratory Standards Institute (CLSI).

Biochemical Analysis

For serum biochemical profiling, the following parameters were evaluated using an automated biochemical analyzer:

- Alanine Aminotransferase (ALT) and Aspartate Aminotransferase (AST) – to assess liver function
- Serum Creatinine and Blood Urea Nitrogen (BUN/ Urea) – to evaluate renal function
- Total Serum Protein – as an indicator of overall metabolic and immune status

All reagents used were of analytical grade, and commercially available diagnostic kits were employed as per manufacturer instructions.

Statistical Analysis

All collected data were compiled and subjected to descriptive statistical analysis. The mean values of hematological and biochemical parameters between the two groups were compared using the Student's t-test to determine statistical significance. Results were considered significant at a p-value < 0.05. Data analysis was performed using SPSS (version XX) or GraphPad Prism, ensuring the robustness of statistical interpretation.

RESULTS

Cattle in the exposed group (n=30), which grazed near pesticide-treated agricultural fields, exhibited several

signs of physiological stress and mild toxicity when compared to the control group (n=30). Common symptoms recorded included:

- Excessive salivation (observed in 40% of exposed cattle)
- Intermittent diarrhea with semi-formed or watery feces (35%)
- Rough and dull hair coat with signs of skin irritation or minor lesions (30%)
- Reduced milk yield in lactating cows (average reduction of ~15–20%)
- Lethargy and reduced appetite, particularly within 24–72 hours of field exposure

No such symptoms were consistently observed in the control group, which remained clinically healthy throughout the study period.

Hematological Findings

Statistical analysis revealed significant differences ($p < 0.05$) in multiple hematological parameters between the two groups:

Parameter	Control Group (Mean ± SD)	Exposed Group (Mean ± SD)	Significance
Hemoglobin (g/dL)	11.8 ± 0.6	9.9 ± 0.8	**p < 0.01**
RBC count ($\times 10^6/\mu L$)	6.5 ± 0.3	5.3 ± 0.4	**p < 0.01**
WBC count ($\times 10^3/\mu L$)	7.2 ± 0.5	9.1 ± 0.6	**p < 0.01**
PCV (%)	35.4 ± 2.1	30.2 ± 1.8	**p < 0.05**

Cattle in the exposed group showed lower hemoglobin levels, RBC count, and PCV, suggesting mild anemia and reduced oxygen-carrying capacity. In contrast, WBC counts were significantly elevated, indicative of an ongoing inflammatory or immunological response, likely triggered by chronic exposure to environmental toxins.

Biochemical Findings

Marked alterations in biochemical markers were also observed in the pesticide-exposed group, reflecting hepatic and renal stress:

Parameter	Control Group (Mean ± SD)	Exposed Group (Mean ± SD)	Significance
ALT (U/L)	32.5 ± 4.1	58.7 ± 6.2	**p < 0.01**
AST (U/L)	72.4 ± 5.6	109.3 ± 7.4	**p < 0.01**
Serum Creatinine (mg/dL)	1.2 ± 0.1	1.8 ± 0.2	**p < 0.01**
Urea (mg/dL)	28.9 ± 3.2	43.5 ± 4.0	**p < 0.01**
Total Protein (g/dL)	7.1 ± 0.4	6.2 ± 0.5	**p < 0.05**

Significantly elevated ALT and AST levels in the exposed group pointed toward hepatic dysfunction, possibly due to the liver's role in detoxifying xenobiotics such as pesticides. Higher creatinine and urea levels were indicative of early signs of renal stress or compromised kidney function. A modest reduction in total protein levels suggested metabolic or nutritional stress, potentially resulting from reduced feed intake or impaired protein synthesis.

DISCUSSION

The results of the present study provide compelling evidence that chronic or repeated exposure to environmental pesticides has a detrimental impact on the overall health of cattle, especially those reared under free-grazing systems in close proximity to intensively farmed, chemically treated agricultural land. Among the pesticide groups, organophosphates emerged as particularly harmful due to their mechanism of action—they inhibit the enzyme acetylcholinesterase, which is essential for normal neurotransmission. This inhibition leads to the accumulation of acetylcholine at synaptic junctions, thereby causing overstimulation of the nervous system. Clinically, this manifests as a range of neurotoxic symptoms, including excessive salivation (hypersalivation), labored or irregular breathing, muscle tremors, and in severe cases, convulsions and paralysis. Biochemical analysis further corroborated these clinical findings.[5-7] Elevated levels of liver enzymes such as alanine aminotransferase (ALT), aspartate aminotransferase (AST), and alkaline phosphatase (ALP) were recorded in exposed animals, strongly indicating hepatic stress. This hepatic disturbance likely results from the liver's attempt to metabolize and detoxify the accumulated pesticide residues, leading to cellular damage and enzyme leakage into the bloodstream. These findings are consistent with those of Patil *et al.* (2017)[8], who reported similar hepatic enzyme elevations in buffaloes exposed to chlorpyrifos in pesticide-treated agricultural zones of Maharashtra. Moreover, hematological profiles revealed a noticeable reduction in hemoglobin concentration and red blood cell (RBC) counts in exposed cattle. Such hematological alterations suggest that pesticide residues may be interfering with erythropoiesis—the process of red blood cell formation—or may be inducing oxidative stress, thereby promoting hemolysis or damage to existing erythrocytes. Pesticide-induced oxidative stress can lead to the generation of reactive oxygen species (ROS), which

damage cellular membranes, including those of red blood cells, contributing to anemia and related clinical symptoms. The clinical manifestations observed in the field—such as general lethargy, reduced feed intake, decreased milk production, and signs of respiratory or neurological distress—were in strong agreement with the biochemical and hematological parameters measured. Together, these findings establish a clear association between environmental exposure to pesticides and the onset of subclinical or chronic toxicity in cattle populations residing near heavily treated agricultural regions.

CONCLUSION

This study provides compelling evidence that free-grazing cattle in pesticide-intensive agricultural zones such as Saharsa district are at significant risk of physiological and biochemical impairments due to chronic exposure to chemical pesticides. Cattle exposed to organophosphates, carbamates, and pyrethroids exhibited clear clinical symptoms—such as excessive salivation, diarrhea, and reduced milk yield—along with hematological alterations like anemia and leukocytosis, and biochemical indicators of liver and kidney dysfunction.

The observed elevation in hepatic enzymes (ALT, AST) and renal markers (urea, creatinine) among exposed animals points to systemic toxicity, potentially resulting from the ingestion and dermal absorption of pesticide residues present in fodder, water, and soil. The reduction in total serum protein further reflects metabolic and nutritional stress.

These findings highlight the often-overlooked collateral effects of pesticide use on non-target animals and underscore the urgent need for integrated pest management practices that minimize environmental contamination. It is recommended that farmers be made aware of the risks associated with open grazing near treated fields and that livestock management policies be updated to include protective measures against agrochemical exposure. Further research is warranted to assess long-term reproductive and neurological impacts and to explore detoxification strategies or dietary interventions that can mitigate pesticide toxicity in livestock.

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